

End-only + Mylar-wrapped EJ-230 bar:
Optical simulation study
Geant4 11.4.0 / SSLG4-OPSim

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Motivation & scope

- ▶ **Controlled optical study:** readout *only* at the two ends ($\pm X$ faces); all other faces wrapped in Mylar. Not a final detector configuration.
- ▶ **Goals:**
 - ▶ Isolate end-readout optical response.
 - ▶ Characterize multi-component attenuation with Mylar.
 - ▶ Benchmark timing estimators in a clean geometry.
- ▶ 31 beam positions ($x = -690 \dots +690$ mm), 2,000 μ^- events per position.

This study

End-only + Mylar = extreme limit. Photons either exit to an end SiPM, **escape** through the bar boundaries, or are absorbed (surface or bulk) after multiple reflections.

No TB counterpart

This configuration has **no direct test-beam counterpart** (April-2026 TB used top readout + Tyvek).

Electronic jitter not included

SPTR (≈ 106 ps) and FastIC (≈ 10 ps) are post-processing steps, not folded into σ_{eq} shown here.

Geometry & simulation setup

- ▶ **Bar:** EJ-230 (OPSC-106), $1400 \times 60 \times 10 \text{ mm}^3$; long axis X .
- ▶ **End-left SiPMs** (IDs 0–7): $-X$ face.
End-right SiPMs (IDs 8–15): $+X$ face.
- ▶ **Wrapping:** Mylar dielectric surface on $\pm Y, \pm Z$. $R = 0.90$, specular lobe = 1.0, $\sigma_\alpha = 0.1^\circ$.
- ▶ **SUM4:** leading-edge at 4 PE; groups $\{0-3\}/\{4-7\}$ (left), $\{8-11\}/\{12-15\}$ (right).
- ▶ **Rayleigh scattering:** active ($\lambda_{\text{Rayl}} \approx 150 \text{ cm}$ at peak; $\propto \lambda^4$; Materials.cc:45–55).

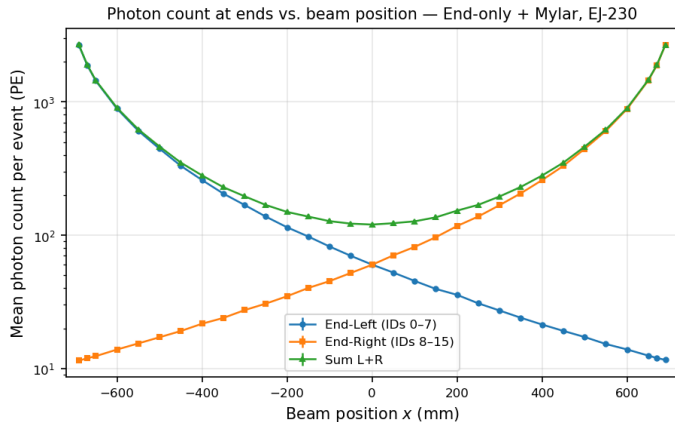
Key parameters

Scint. yield	9,700 γ/MeV
Emission peak	391 nm
Bulk λ (ABSLENGTH)	120 cm
n (flat)	1.58
$v_g = c/n$	189.7 mm/ns
τ_r/τ_d	0.5 ns/1.5 ns
Mylar R	0.90

Note on v_g

No GROUPVEL property set. Geant4 computes $v_g = c/n = 189.7 \text{ mm/ns}$ from flat $n = 1.58$.

Photon yield at ends vs. beam position



Representative values

Near **left** end ($x = -690$ mm):

End-Left (near) ≈ 2675 PE

End-Right (far) ≈ 11.4 PE

Center ($x = 0$, center-point):

End-Left ≈ 60.1 PE

Central avg ($|x| \leq 100$ mm):
 ≈ 62.0 PE

Near **right** end ($x = +690$ mm):

End-Right (near) ≈ 2667 PE

End-Left (far) ≈ 11.6 PE

Two-scale structure

Near-end: prompt contribution.

Center: ≈ 62.0 PE residual
sustained by Mylar recirculation
(many bounces at $R = 0.90$).

Bulk attenuation and longitudinal light-yield scales

ABSLENGTH = 120 cm (material property)

Describes absorption along the **actual optical path length** s : $P_{\text{bulk}}(s) = e^{-s/\lambda_{\text{bulk}}}$. A photon propagating 70 cm longitudinally may traverse significantly more path via reflections.

ABSLENGTH knows nothing about bar geometry or reflectors.

Fitted longitudinal scale (system observable)

NPE vs. distance d combines: bulk absorption (over actual s), Rayleigh scattering, geometry, repeated Mylar surface losses, SiPM acceptance, and PDE. **Any fitted scale is a system-dependent empirical parameter.**

Why single-exp fails

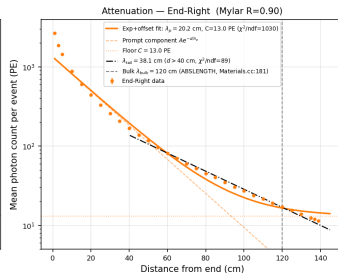
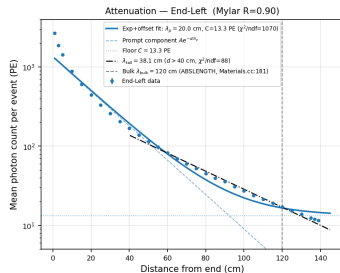
Three numbers inconsistent with $N_0 e^{-d/\lambda}$:

1. Two-point: $\lambda \approx 19$ cm
2. Fit: $N_0 = 782$ PE, $\lambda = 29.2 \pm 1.6$ cm
3. Observed at $d \approx 0$: 2675 PE

Items (2) and (3) disagree by $\approx 3.6\times$.
 $\chi^2/\text{ndf} = 2367$: catastrophic. **Cause:**

superposition of a prompt component and a long-range contribution with finite-range offset.

Attenuation: four models and empirical tail slope



M3 result (exp + finite-range offset)

$$N(d) = A e^{-d/\lambda_p} + C$$

$\lambda_p = 20.0 \pm 1.0$ cm (prompt scale)

$C = 13.3 \pm 1.6$ PE (finite-range offset)

$$\chi^2/\text{ndf} = 1070$$

M4 (empirical tail slope)

$d > 40$ cm:

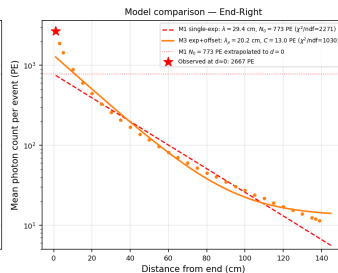
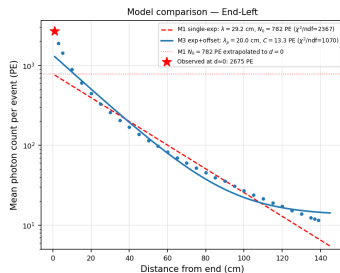
$$\lambda_{\text{tail}} = 38.1 \pm 0.8$$
 cm

$$\chi^2/\text{ndf} = 87.8$$

All $\chi^2/\text{ndf} \gg 1$

All models are **empirical approximations**. λ_p , λ_{tail} are **system-dependent scales**, not the EJ-230 bulk ABSLENGTH (120 cm).

Model comparison: M1, M2, M3, M4 (left side)



Fit diagnostics (covariance errors)

Parameter	Value	χ^2/ndf
M1: λ	29.2 ± 1.6 cm	2367
M3: λ_p	20.0 ± 1.0 cm	1070
M3: C (offset)	13.3 ± 1.6 PE	—
M2: A_s	2104	101
M2: λ_s	9.0 ± 0.4 cm	—
M2: A_I	373	—
M2: λ_I	38.9 ± 0.8 cm	—
M4: λ_{tail}	38.1 ± 0.8 cm	87.8

Empirical approximations

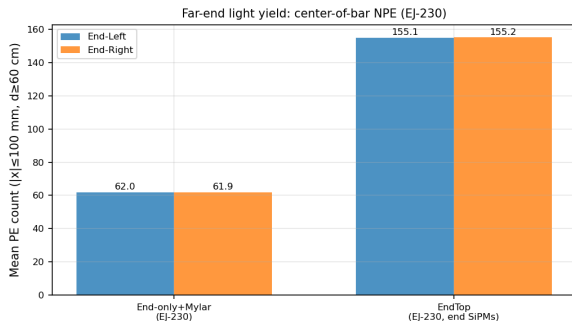
Red star: observed NPE

≈ 2675 PE at $d \approx 0$.

M1 $N_0 = 782$ PE: factor ≈ 3.6 off.

M2 identifies two empirical scales but $\chi^2/\text{ndf} = 101 \gg 1$: not an exact description.

Far-end light yield: where reflector quality matters



Central-region NPE ($|x| \leq 100$ mm)

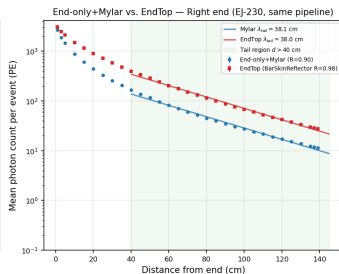
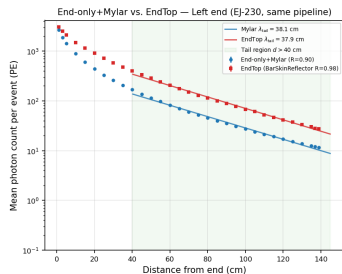
Config	R	NPE
End-only+Mylar	0.90	62.0 PE
EndTop	0.98	155.1 PE

Ratio EndTop/Mylar $\approx 2.5\times$.

Interpretation

Both see photons after ≥ 60 cm and many reflections. Mylar loss/interaction: $1-R = 0.10$ (10%). Accumulated over many bounces, this strongly suppresses far-end yield. $2.5\times$ difference is **consistent with** higher- R boundaries in EndTop ($R = 0.98$), but this is a **configuration-level result**.

Apples-to-apples: End-only+Mylar vs. EndTop (end SiPMs only)



Identical pipeline

Both datasets: same end-SiPM-only selection, same models, same tail-fit range $d > 40$ cm.

Empirical tail slope ($d > 40$ cm)

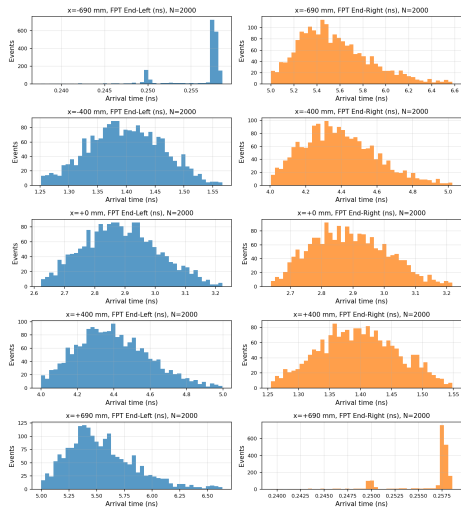
Config	λ_{tail}
End-only+Mylar	38.1 ± 0.8 cm
EndTop (end SiPMs)	37.9 ± 0.6 cm

Numerically similar slopes in this fit range. Does *not* establish reflector independence or exclude model error.

Key conclusion

NPE amplitude at center differs $2.5\times$. The tail slope is not the discriminating observable; the absolute light yield is. Difference is consistent with higher- R boundaries (configuration-level).

Arrival-time distributions: FPT at representative positions



Distributions shown

FPT at End-Left and End-Right.

$x \in$

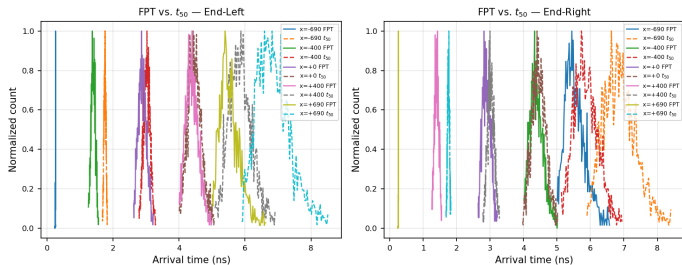
$\{-690, -400, 0, +400, +690\}$ mm.

$N = 2,000$ events/position.

Shape

Distribution shifts later and broadens as beam moves from the SiPM: longer path, fewer photons, estimator-dependent response to photon-population size.

FPT vs. t_{50} : estimator comparison



FPT vs. t_{50}

FPT: minimum arrival time per event.

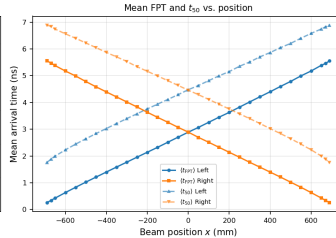
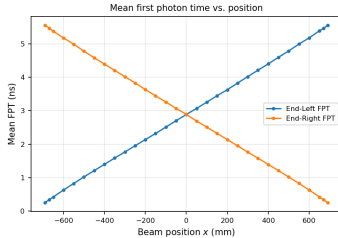
t_{50} : per-event median — less sensitive to photon-population size effects. t_{50} is systematically

later than FPT; offset grows with distance.

Estimator-dependent response

Both FPT and t_{50} slopes exhibit position-dependent apparent velocities. These reflect the estimator-dependent apparent longitudinal response: systematic shifts of the estimator mean with photon-population size and reflection path-length distribution. This is a sampling and bias effect, not superluminal transport.

Position dependence of timing estimators



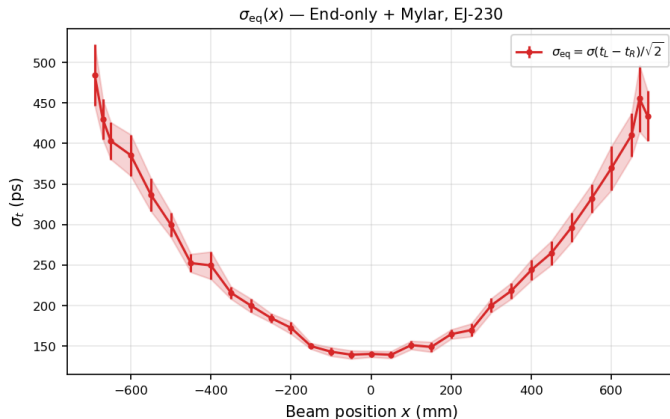
Apparent estimator velocities

Estimator	v_{app}
SUM4 Δt	210.3 mm/ns
FPT slope	263.6 mm/ns
t_{50} slope	275.5 mm/ns
c/n (theory)	189.7 mm/ns

All exceed c/n

These are estimator-dependent apparent longitudinal responses, not direct measurements of photon group velocity. All exceed $c/n = 189.7 \text{ mm/ns}$ ($n = 1.58$, flat RINDEX). Reflects position-dependent photon-population and estimator biases (reflections, path-length distributions, threshold selection).

Intrinsic timing metric $\sigma_{\text{eq}}(x)$



Key values of σ_{eq}

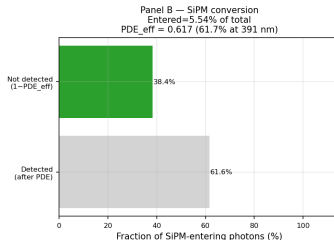
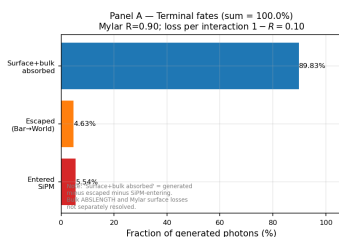
x (mm)	σ_{eq} (ps)
-690	485 ± 38 ps
-400	249 ± 17 ps
0	140 ± 4 ps
+400	244 ± 13 ps
+690	434 ± 31 ps

Definition and caveats

$\sigma_{\text{eq}} \equiv \sigma(t_L - t_R)/\sqrt{2}$ (SUM4, core Gaussian fit). **Equals per-end resolution only under equal, independent-end assumptions** — which break down at the bar ends where NPE is strongly asymmetric. Intrinsic

optical+estimator only. SPTR and electronics **not** included.

Photon budget — End-only + Mylar



Panel A — Terminal fates

Fate	Fraction
Surface+bulk absorbed	≈ 89.8%
Escaped	4.63%
Entered SiPM	5.54%
Total	≈ 100%

Panel B — SiPM conversion

$$\epsilon_{PDE,eff} = N_{det}/N_{SiPM} \approx 0.62$$

Loss note

Surface and bulk absorption are **not** separately resolved: the census cannot distinguish which mechanism removed the photon.
Mylar loss/interaction:
 $1-R=0.10$.

Summary

Key findings

- ▶ **ABSLENGTH** = 120 cm = material property (actual path, not longitudinal distance). All fitted scales are system-dependent.
- ▶ **Tail slope** ($d > 40$ cm): $\lambda_{\text{tail}} = 38.2$ [37.2, 39.2] cm (bootstrap 68 % CI, 200 replicas).
- ▶ **M1**: $\chi^2/\text{ndf} = 2367$. **M2**: two scales, $\chi^2/\text{ndf} = 101 \gg 1$.
- ▶ **EndTop** ($R = 0.98$): $2.5\times$ more central NPE. Consistent with higher- R ; configuration-level result.
- ▶ v_{app} all exceed $c/n = 189.7$ mm/ns: estimator-dependent apparent responses.
- ▶ $\sigma_{\text{eq}} = 140 \pm 4$ ps (center), intrinsic only. **Add SPTR+electronics separately.**

Next steps

1. $\hat{t}_0 - t_0^{\text{true}}$ as primary metric.
2. SPTR *per detected photon*; no quadrature sum.
3. Separate surface from bulk in census.
4. Reflectivity scan ($R = 0.85\text{--}0.95$).
5. Simulate the actual TB geometry.

Appendix A: simulation parameters

Parameter	Value	Unit	Source
Bar material	EJ-230 (OPSC-106)		src/Materials.cc:163
Bar dimensions	$1400 \times 60 \times 10$	mm ³	GDML
Readout	End-only ($\pm X$ faces)		geometry
Wrapping ($\pm Y, \pm Z$)	Mylar surface (dielectric_metal)		src/DetectorConstruction.cc:236
Mylar reflectivity R	0.90		src/Materials.cc:347 (CreateMylarReflector)
Mylar specular lobe	1.0		src/Materials.cc:348
Mylar σ_α	0.1	deg	src/Materials.cc:344
Loss per interaction $1 - R$	0.10		derived; $R = 0.90$
Scint. yield	9,700	γ/MeV	src/Materials.cc:182
Emission peak	391	nm	src/Materials.cc:181
Bulk λ (ABSLENGTH)	120	cm	src/Materials.cc:181; DetectorConstruction.cc:183-186
Refractive index n	1.58		src/Materials.cc:41 (flat, 4 energy points)
Density	1.023	g/cm ³	src/Materials.cc:24 (PVT)
Rise time τ_r	0.5	ns	src/Materials.cc:182; DetectorConstruction.cc:194
Decay time τ_d	1.5	ns	src/Materials.cc:182
SiPM (physical)	AFBR-S4N66P014M		hardware
SiPM PDE (at 391 nm)	0.6165		interpolated: data/sipm/AFBR-S4N66P024M_pde.txt
Beam	$\mu^-, 1 \text{ GeV}$		macro
Positions	31		run_metadata
Events/position	2,000		run_metadata

Appendix B: σ_{eq} per-position summary

x (mm)	N_{trig}	Trig. eff.	σ_{eq} (ps)	Err. (ps)
-690	1564	0.782	484.6	37.7
-400	1988	0.994	249.5	16.9
+0	2000	1.000	140.2	3.8
+400	1989	0.995	243.9	12.9
+690	1569	0.784	434.0	30.7

$\sigma_{\text{eq}} = \sigma(t_L - t_R)/\sqrt{2}$ from SUM4 leading-edge core Gaussian fit. Equals per-end resolution only under equal, independent-end assumptions. Intrinsic optical+estimator only; SPTR and electronics not included.

Appendix C: attenuation fit details (four models)

Model	Side	N_0 or A (PE)	λ (cm)	C (PE)	χ^2/ndf
Single-exp (M1)	Left	782 ± 100	29.2 ± 1.6	—	2367
Exp+offset (M3)	Left	1341 ± 132	20.0 ± 1.0	13.3 ± 1.6	1070
λ_{tail} M4 ($d > 40$ cm)	Left	394 ± 19	38.1 ± 0.8	—	87.8
Single-exp (M1)	Right	773 ± 99	29.4 ± 1.6	—	2271
Exp+offset (M3)	Right	1320 ± 130	20.2 ± 1.0	13.0 ± 1.6	1030
λ_{tail} M4 ($d > 40$ cm)	Right	393 ± 19	38.1 ± 0.8	—	88.6

Weighted fits (per-point SEM). Errors from covariance diagonal. M4 tail: $d > 40$ cm. All $\chi^2/\text{ndf} \gg 1$: empirical approximations only.

Appendix D: empirical tail-slope range sensitivity

$d_{\min}$	$\lambda_{\text{tail}}^{\text{L}}$	$\lambda_{\text{tail}}^{\text{R}}$	χ^2/ndf (L)
30 cm	36.4 ± 0.8 cm	36.5 ± 0.8 cm	170
40 cm	38.1 ± 0.8 cm	38.1 ± 0.8 cm	88
50 cm	39.7 ± 0.7 cm	39.7 ± 0.7 cm	45

The empirical tail slope increases with $d_{\min}$, indicating residual curvature even in the “tail” region.
Reference: $d_{\min} = 40$ cm.

Appendix E: M2 and M4 bootstrap stability (200 replicas, left side)

Parameter	Median	P16	P84
M2: λ_s (cm)	9.1	8.2	10.2
M2: λ_l (cm)	39.2	37.9	40.8
M4: λ_{tail} (cm)	38.2	37.2	39.2

Method: position-level pairs bootstrap, C++/OpenMP, seed 230123, 200 replicas. M2 failure rate: 0%. Intervals are 68 % bootstrap CIs (P16–P84), distinct from covariance errors reported in Appendix C. λ_l^{M2} consistent with $\lambda_{\text{tail}}^{\text{M4}}$.

Appendix F: group-velocity and slope-velocity discussion

Theoretical $v_g = c/n$

Flat $n = 1.58$ in `Materials.cc:41`. No explicit GROUPVEL property set (Geant4 computes $v_g = c/n$):

$$v_g = 299.792/1.58 = 189.7 \text{ mm/ns}.$$

Apparent slope velocities

Estimator	v_{app}
SUM4 Δt	210.3 mm/ns
FPT	263.6 mm/ns
t_{50}	275.5 mm/ns
c/n (theory)	189.7 mm/ns

Why all exceed c/n

Slope-derived velocities are **estimator-dependent apparent longitudinal responses**. SUM4 Δt :

$v_{\text{app}} = 2/|d\Delta t/dx|$. Photons travel longer actual paths via reflections, delaying both SUM4 times; net Δt slope is shallower than a direct-path signal would give. FPT and t_{50} : both shift

systematically with photon-population size and reflection path-length distribution, yielding estimator-specific biases. An axial single-photon injection test (no reflections, known d) is needed to validate Geant4's v_g .

Appendix G: method & caveats

- ▶ **Bar geometry:** $1400 \times 60 \times 10 \text{ mm}^3$, axis X . End SiPMs on $\pm X$; Mylar dielectric surface on $\pm Y, \pm Z$.
- ▶ $\sigma_{\text{eq}} = \sigma(t_L - t_R)/\sqrt{2}$ is an *equivalent* per-end metric assuming equal, independent ends. Not a bar-level resolution measurement.
- ▶ **Intrinsic only:** SPTR ($\approx 106 \text{ ps}$) and FastIC ($\approx 10 \text{ ps}$) not folded in. Apply SPTR at the photoelectron level; do not add full SPTR directly in quadrature to event resolution.
- ▶ **SiPM naming:** AFBR-S4N66P024M used only for the PDE curve; physical device is AFBR-S4N66P014M.
- ▶ **Photon budget:** surface+bulk absorption not separately resolved in the boundary census.
- ▶ **Rayleigh:** active ($\lambda_{\text{Rayl}} \approx 150 \text{ cm}$ at 391 nm , $\propto \lambda^4$); constitutes an additional scattering mechanism beyond ABSLENGTH.
- ▶ **No TB counterpart:** April-2026 TB used top readout + Tyvek $\neq$ end-only + Mylar.
- ▶ **EndTop reference:** EXEC_07 data, 2,000 events/position, 31 positions. Side reflectors $R = 0.98$ (BarSkinReflector).
- ▶ **Bootstrap:** position-level pairs resampling, C++/OpenMP, seed 230123. 68 % CI = P16–P84 interval; distinct from covariance diagonal errors reported elsewhere.